Lab Assignment 2

Docker Image Build and Publishing with GitHub Actions

CSC418 DevOps

Reuse the same application source code and Dockerfile that you created in Lab Assignment 1. Automate the build and publication of that application using two GitHub Actions workflows: one with Docker commands and one with prebuilt actions. Publish both resulting images to your Docker Hub account.

Required image names

Replace xx with your two-digit registration numbe. Create two public Docker Hub repositories and publish the following image tags:

`<dockerhub_username>/bse_xx_commands:latest`

`<dockerhub_username>/bse_xx_actions:latest`

For example, registration number 07 uses bse_07_commands and bse_07_actions. Both must build the same application from the same Dockerfile and source revision; only the build and publishing method changes.

Part 1 Prepare the repository and credentials

- Create or reuse a GitHub repository containing your Lab Assignment 1 source, Dockerfile, and all files needed by the Docker build context. Keep the application entry point and port configuration.
- Create the folder .github/workflows and use two separate workflow files: docker-commands.yml and docker-actions.yml.
- Create a Docker Hub personal access token with permission to read and write images. In GitHub repository Settings > Secrets and variables > Actions, add DOCKERHUB_USERNAME and DOCKERHUB_TOKEN as repository secrets.
- Configure both workflows to run on pushes to your default branch and manually with workflow_dispatch. Use a GitHub-hosted Linux runner such as ubuntu-latest. Grant the GitHub token only contents: read permission.
- Keep credentials out of source files, workflow text and screenshots. Read them through GitHub Actions secrets; do not print the token in logs.

Part 2 Build and publish using Docker commands

In .github/workflows/docker-commands.yml:

- Check out the repository. Using actions/checkout for this step is allowed.
- Use run steps to sign in with docker login, build with docker build, and publish with docker push. Supply the token to docker login through --password-stdin using a secret-backed environment variable.
- Build directly from the checked-out Dockerfile and project files. Tag and push the image as <dockerhub_username>/bse_xx_commands:latest.
- Perform Docker authentication, building and publishing through shell commands. Do not use docker/login-action or docker/build-push-action in this workflow.

Part 3 Build and publish using prebuilt actions

In .github/workflows/docker-actions.yml:

- Use actions/checkout to obtain the same project files.
- Use docker/login-action to authenticate with the same repository secrets and docker/setup-buildx-action to configure the builder.
- Use docker/build-push-action to build and publish the image. Set the correct context and Dockerfile path, enable push: true, and tag the result as <dockerhub_username>/bse_xx_actions:latest.
- Perform Docker authentication, building and publishing through these prebuilt actions, rather than shell commands.

Part 4 Verify publication

- Run both workflows successfully for the same source commit. Inspect their logs to confirm that each build and push completed.
- Open both public Docker Hub repositories and confirm that the latest tag is present. Record each image name, tag and repository URL.

Submission requirements

Submit one report containing:

- Your name and registration number, the GitHub repository URL, and the source commit used for both successful runs. Ensure the instructor can access the repository and workflow runs.
- The complete Dockerfile and both workflow YAML files as selectable text, with their filenames clearly identified.
- The public URLs of both Docker Hub repositories and the full image names with tags.
- A link and screenshot for each successful workflow run. Show the relevant build and push steps, not only the workflow list.
- Docker Hub screenshots showing the latest tag in each of the two repositories. Do not expose credentials.
- A brief comparison of the command-based and prebuilt-action approaches, explaining what run and uses do and how credentials are handled.